ME Machine Lab

Detailed Step-by-Step Assembly Manuals

Eight catalog-parts STEM outreach projects

Important release status

Every project in this manual is a procurement-complete rendered concept. It is not yet a classroom-release design. A physical first article must verify all interfaces, motion, containment, and the project-specific safety boundary before students operate it.

No-fabrication rule

Students and facilitators may select, position, thread, fasten, tie, route, and adjust catalog parts through their existing features. They may not alter a purchased part. Cutting, drilling, machining, sanding, bending, heat-forming, welding, soldering, or 3D-printing a build component is outside this manual.

How to use this manual

Read the project safety boundary first, inventory exact catalog parts, assemble in sequence, verify each step by hand, then complete the first-article release checks. The detailed product links and price information are maintained in the design README files and consolidated purchase list because vendor availability changes.

Included projects

- Cable-Car Dash: Build a counterweighted gondola, then tune payload and stopping error.
- Crank Critter: Build a stationary two-cam automaton and investigate timing, follower stroke, and gear reduction.
- Gear Garden: Make a guarded hand-cranked gear sculpture and predict speed and direction changes.
- HydroGrab: Build a water-actuated arm and compare predicted with observed force and displacement.
- Build-a-Monocular Lab: Assemble a 3x Galilean monocular and test focus and angular magnification using distant terrestrial targets.
- Pinball Lab: Build a reconfigurable tabletop pinball game and quantify how layout changes affect ball paths and game time.
- Rescue Hoist: Build a two-stage 9:1 hand winch and compare torque reduction with input-turn count.
- Springy Wobblebot: Build a low-voltage vibrating creature and investigate how unbalance, compliant legs, and friction produce motion.

Generated from repository design READMEs on 2026-11-04.

Cable-Car Dash: counterweighted tower toy

Grades 6-12 | 60-90 min

DRAFT INSTRUCTION MANUAL - RENDERED CONCEPT; PHYSICAL FIRST ARTICLE REQUIRED BEFORE CLASSROOM USE

Build a counterweighted gondola, then tune payload and stopping error.



1. Purpose and status

This manual describes an exact-catalog, no-fabrication build sequence. The accompanying render communicates intended arrangement only; it does not verify dimensions, fit, strength, containment, or classroom safety.

2. Safety boundary

Hand power only. Keep the counterweight inside the tower-base footprint. Start with no more than a 50 g gondola payload until the first article passes its checks.

Use only factory-provided holes, threads, bores, mounting patterns, and supplied fasteners. Do not cut, drill, punch, mill, saw, bend, heat-form, solder, glue, or 3D-print any component.

3. Before building: inventory and set up

- Choose a level tabletop away from its edge. Mark the counterweight travel envelope and keep hands out of it.
- Separate the guide rope from the drive rope before building. Do not cut, melt, splice, or substitute rope.
- Confirm that each pulley, shaft collar, and tower fastener is present before raising either tower.

Pull-list bill of materials

Use the exact vendor item or ASIN shown below. The canonical README in the matching design folder retains the direct procurement hyperlinks and package-yield notes.

Function	Exact item and purchased configuration	Buy
Towers, bases, trolley plate, counterweight plate, square shafts, bearings, standoffs, #8-32 fastening hardware, and two supplemental angle gussets	VEX V5 276-2161, Metal & Hardware Kit	1 kit
Vertical tower members and base cross-members	VEX V5 276-2288, 1x2x1x25 Aluminum C-Channel	1 pack
Tower/base bracing	VEX V5 276-7761, 90-degree Gusset Pack	1 pack
Travel rollers, hand winch, ratchet, and three pre-made ropes	VEX V5 276-1546, Winch & Pulley Kit	3 kits
Gondola body with factory mounting features	McMaster-Carr 5257N111, gray impact-resistant polycarbonate enclosure, 3.75 x 4.5 x 2.25 in.	1
Nonstructural tethering/line retention	VEX V5 275-0125, 11-in. zip ties (100-pack)	1 pack
Shared assembly tools	VEX V5 276-1382, Tool Kit	1 shared kit

4. Step-by-step assembly

Complete the numbered steps in order. After each mechanical change, move the mechanism by hand with power off and with no payload unless the step explicitly directs otherwise.

Step 1. Use two full 276-2288 C-channels as upright tower members and two as bases. Join each upright to its base with four flat 276-7761 gussets; use the two bent 276-7761 gussets and two 276-2161 angle gussets as additional base braces. Join the towers with two 276-2161 chassis rails. Position the bases parallel on a level tabletop; do not clamp the system near an edge.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 2. Bolt a 276-2161 pillow block and a 276-1546 pulley at the top of each tower using only factory holes. Keep their centers at equal height. Use the supplied VEX hardware and tighten both tower bases before adding rope.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 3. Make the trolley from one supplied 5x15 steel plate, four pulley-kit pulleys, two supplied short shafts, and the supplied shaft collars. Mount the 5257N111 enclosure to the plate through its factory external mounting holes; use spacers from the kit if needed to keep the enclosure clear of the wheels.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 4. Tie one uncut, factory 4-ft rope taut between the tower tops as the *guide* line. Put four pulley-kit pulleys on the trolley plate so the trolley captures and rolls along this guide line. Use only the pulley grooves and factory VEX shaft hardware; do not cut, melt, or splice the rope.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 5. Join the ends of the second and third uncut, factory 4-ft ropes with a double-fisherman's bend to make the *drive* line; do not cut or splice either rope. Attach one free end to the trolley plate, route it over the pulley at one tower, and attach the other free end to the other 5x15 steel plate, which is the single secured counterweight module. Mount the VEX winch at the counterweight tower so the facilitator can reset the counterweight by hand. Keep it inside the space bounded by the tower bases throughout travel.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 6. With the switchless, hand-powered system unloaded, crank slowly through the full travel. Add no more than 50 g payload to the enclosure until the first-article checks are complete.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

5. First-article release checks

A build is not ready for student operation until a facilitator completes and records each applicable check below.

• With the counterweight at either extreme, the tower bases must remain stable and all moving mass must stay inside the tabletop footprint.

• With a 50 g payload, cycle full travel 20 times; inspect pulley retention, rope abrasion, cable ties, and enclosure fasteners after each five cycles.

• Establish and mark a no-hand zone around counterweight travel and around the winch/ratchet. The counterweight must not be a collection of loose washers.

7. Engineering investigation

For the proposed one-to-one rope path, a hanging counterweight provides approximately m_mathrmcwg while the moving inertia is m_mathrmcw+m_mathrmcar+m_mathrmpayload. Neglecting rope and pulley losses, aapprox m_mathrmcwg/(m_mathrmcw+m_mathrmcar+m_mathrmpayload). Students should define a positive travel direction, predict the acceleration, then compare with stopping error at each station.

Closeout

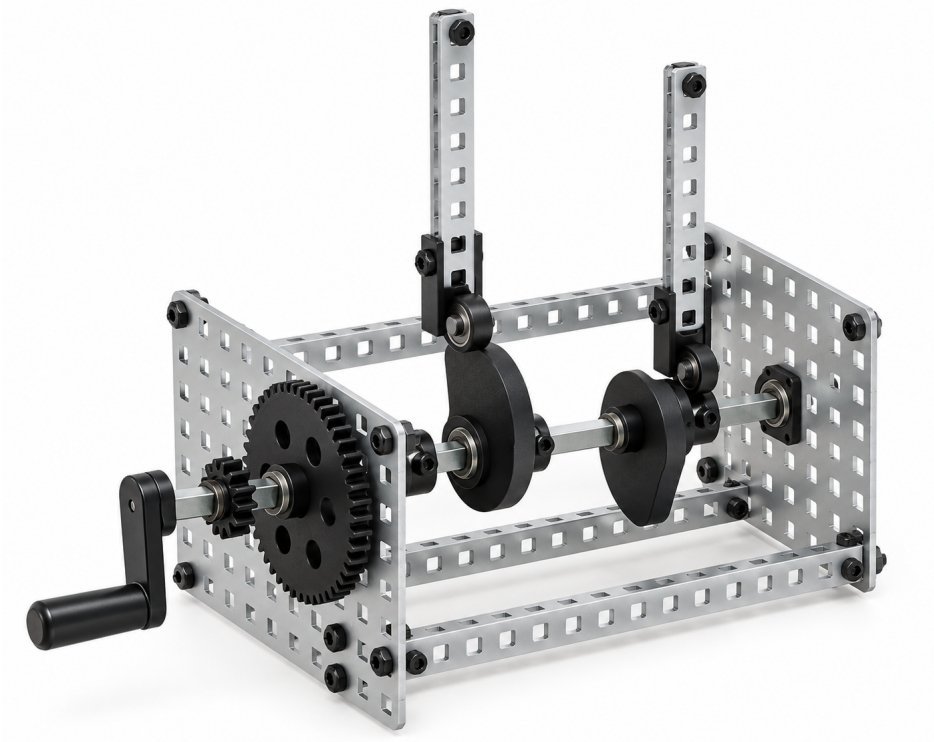
Turn off or unload the mechanism, return reusable components to labeled containers, dry any water system, and record one configuration change plus one observed result. Remove any damaged, loose, frayed, leaking, or hot component from service.

Crank Critter: dual-cam automaton

Grades 5-10 | 45-75 min

DRAFT INSTRUCTION MANUAL - RENDERED CONCEPT; PHYSICAL FIRST ARTICLE REQUIRED BEFORE CLASSROOM USE

Build a stationary two-cam automaton and investigate timing, follower stroke, and gear reduction.



1. Purpose and status

Teams make a hand-cranked tabletop creature/machine whose two cam followers produce repeating, phase-offset motions (for example, a head bob and a tail wave). It is deliberately a stationary automaton, not a walking mechanism: that gives students a buildable, no-fabrication route to timing, cam profile, friction, and gear ratio without unsupported craft parts.

This manual describes an exact-catalog, no-fabrication build sequence. The accompanying render communicates intended arrangement only; it does not verify dimensions, fit, strength, containment, or classroom safety.

2. Safety boundary

Hand crank only. This is a stationary automaton, not a walking machine. Stop before adjusting a cam, follower, gear, or pivot.

Use only factory-provided holes, threads, bores, mounting patterns, and supplied fasteners. Do not cut, drill, punch, mill, saw, bend, heat-form, solder, glue, or 3D-print any component.

3. Before building: inventory and set up

• Sort the frame hardware, bearings, collars, shafts, 12T gear, 36T gear, cams, followers, and hand crank into separate groups.

• Use the VEX 24-DP gear pair only; do not combine gears from different pitch systems.

• Keep the frame on a flat table. The base must sit without rocking before a crank is installed.

Pull-list bill of materials

Use the exact vendor item or ASIN shown below. The canonical README in the matching design folder retains the direct procurement hyperlinks and package-yield notes.

Item	Exact purchased configuration and use	Qty. used / purchased
Frame and fasteners	VEX Robotics 276-2161 Metal & Hardware Kit; use two 5x15 plates, four 1x25 bars, supplied #8-32 fasteners, bearings, collars, and short square shafts	1 / 1 kit
Crank, cams, and followers	VEX Robotics 276-2045 Advanced Mechanics and Motion Kit; use one hand crank, two cams, and two cam followers. The one consolidated kit supplies this project and its second hand crank is allocated to Rescue Hoist.	1 / 1 shared kit
Speed-reduction gears	VEX Robotics 276-2169 Gear Kit; use one 12T and one 36T 24-DP, 20° gear	2 / 1 kit
Shared hand tools	VEX Robotics 276-1382 Tool Kit; use the supplied hex key/wrench with the kit hardware	shared / 1 kit

4. Step-by-step assembly

Complete the numbered steps in order. After each mechanical change, move the mechanism by hand with power off and with no payload unless the step explicitly directs otherwise.

5. First-article release checks

A build is not ready for student operation until a facilitator completes and records each applicable check below.

- The 12T/36T pair is from the same 276-2169 24-DP, 20° gear kit and runs without climbing/skipping at hand speed.
- Both followers remain in their factory-hole guides through ten full crank cycles.
- All frame/shaft fasteners are from the #8-32 276-2161 family; no improvised pivot hardware is used.
- The automaton stays on the table without a person holding its base.

6. Troubleshooting

Observation	Likely physical cause	First correction
Crank jams	Gear mesh or bearing alignment is too tight	Shift one shaft by an existing hole position; confirm shafts are parallel
Follower loses contact	Guide position/height is wrong	Reposition guide using a factory hole; retain follower before retrying
Motion is too small	Cam orientation gives little visible follower travel	Reposition a follower guide using another existing hole
Frame rocks	Uneven base or loose crossmember	Retighten base hardware and choose a flat work surface

Closeout

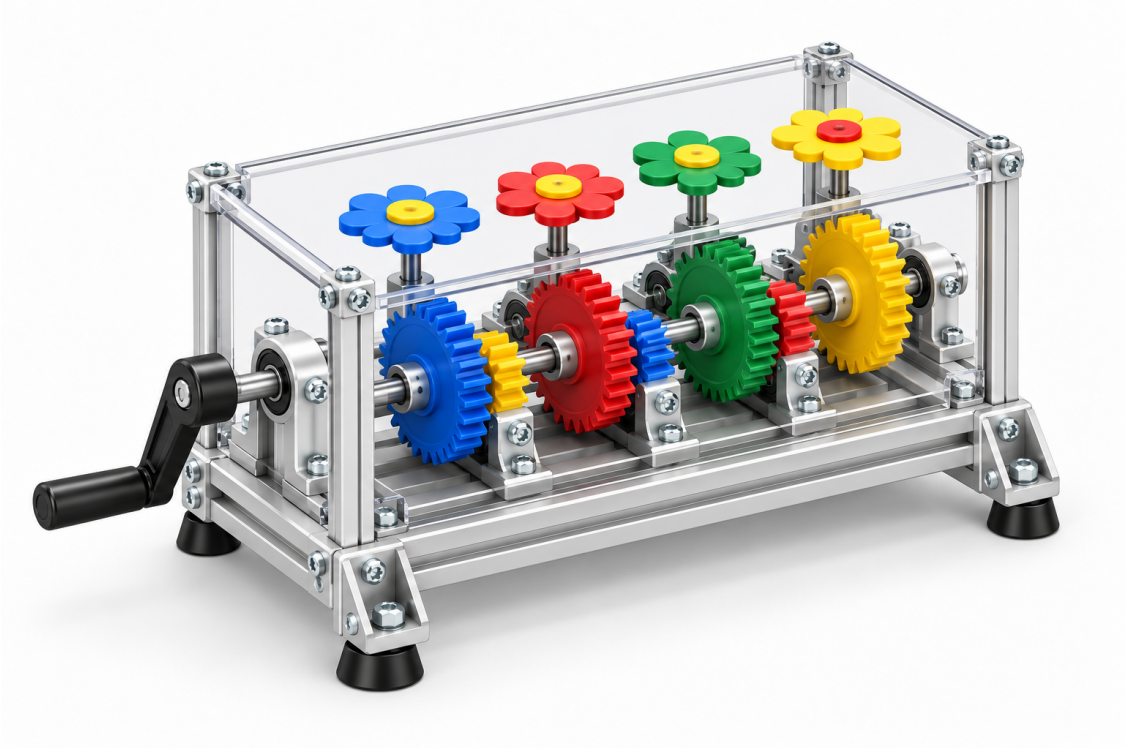
Turn off or unload the mechanism, return reusable components to labeled containers, dry any water system, and record one configuration change plus one observed result. Remove any damaged, loose, frayed, leaking, or hot component from service.

Gear Garden: hand-cranked kinetic sculpture

Grades 5-12 | 45-75 min

DRAFT INSTRUCTION MANUAL - RENDERED CONCEPT; PHYSICAL FIRST ARTICLE REQUIRED BEFORE CLASSROOM USE

Make a guarded hand-cranked gear sculpture and predict speed and direction changes.



1. Purpose and status

This manual describes an exact-catalog, no-fabrication build sequence. The accompanying render communicates intended arrangement only; it does not verify dimensions, fit, strength, containment, or classroom safety.

2. Safety boundary

Turn only by the prescribed hand crank. The upper plate is a finger barrier, not decoration; do not operate if any student can reach a gear mesh.

Use only factory-provided holes, threads, bores, mounting patterns, and supplied fasteners. Do not cut, drill, punch, mill, saw, bend, heat-form, solder, glue, or 3D-print any component.

3. Before building: inventory and set up

- Pull exactly two perforated plates and eight matching standoffs; do not replace them with cut sheet or improvised spacers.
- Lay out the 12T, 36T, 60T, and 84T gears. Confirm each has the VEX square-shaft interface before assembly.
- Keep cable ties outside the plate sandwich and untrimmed. They are visual indicators only, never load-bearing members.

Pull-list bill of materials

Use the exact vendor item or ASIN shown below. The canonical README in the matching design folder retains the direct procurement hyperlinks and package-yield notes.

Function	Exact item and purchased configuration	Buy
Perforated base and fixed upper finger guard	VEX V5 276-2311, 5 x 25 Aluminum Plate	1 pack
Shafts, flat bearings, shaft collars, standoffs, washers, #8-32 screws and nuts; one factory-hole 1x25 crank bar	VEX V5 276-2161, Metal & Hardware Kit	1 kit
Matched 24-DP gears	VEX V5 276-2169, Gear Kit	1 kit
Decorations and non-load-bearing labels	VEX V5 276-1032, cable ties	1 pack
Shared assembly tools	VEX V5 276-1382, Tool Kit	1 shared kit

4. Step-by-step assembly

Complete the numbered steps in order. After each mechanical change, move the mechanism by hand with power off and with no payload unless the step explicitly directs otherwise.

Step 1. Place two 276-2311 plates parallel, with the first as the base and the second as the upper guard. Join them with eight supplied 1-in. VEX standoffs at the factory-hole grid, producing a protected internal gear space.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 2. Install four VEX flat bearings in selected factory holes of both plates. Pass the supplied shafts through matched bearing pairs; capture each with supplied shaft collars so no shaft can drift into a plate.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 3. Place the 12T, 36T, 60T, and 84T gears from 276-2169 between the plates. Begin with 12T-to-36T and 36T-to-60T meshes; use the 84T gear only after the first two meshes rotate freely. The teeth must mesh without forced contact.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 4. Lock each driven gear to its shaft with a supplied collar or the gear’s intended shaft engagement. Bolt one 276-2161 1x25 bar to the 60T gear’s factory mounting holes to make the hand crank. Its unused full length is intentional; do not shorten it.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 5. Add up to eight full cable ties as petals/flags only on exterior hardware. Keep every tie end outside the sandwich and at least one gear radius from a mesh; do not trim the tails.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 6. Turn the crank slowly through 20 revolutions. If any mesh binds, stop and reposition the shaft using only factory holes before allowing students to operate it.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

5. First-article release checks

A build is not ready for student operation until a facilitator completes and records each applicable check below.

• Verify the selected gear centers give clearance/backlash rather than tooth-bottom contact; all gears in 276-2169 share the specified 24-DP, 20-degree system.

• Confirm the upper plate and standoff spacing prevent a child’s finger from reaching the faces or teeth while the crank is turned.

• Confirm the bar crank clears the plate, tabletop, nearby students, and its own cable-tie ornaments through a full rotation.

7. Engineering investigation

For two meshing external gears, omega_2/omega_1=-N_1/N_2. The minus sign denotes direction reversal. Have students predict the direction and relative speed of the 36T and 60T gears before turning the crank, then compare prediction with observation.

Closeout

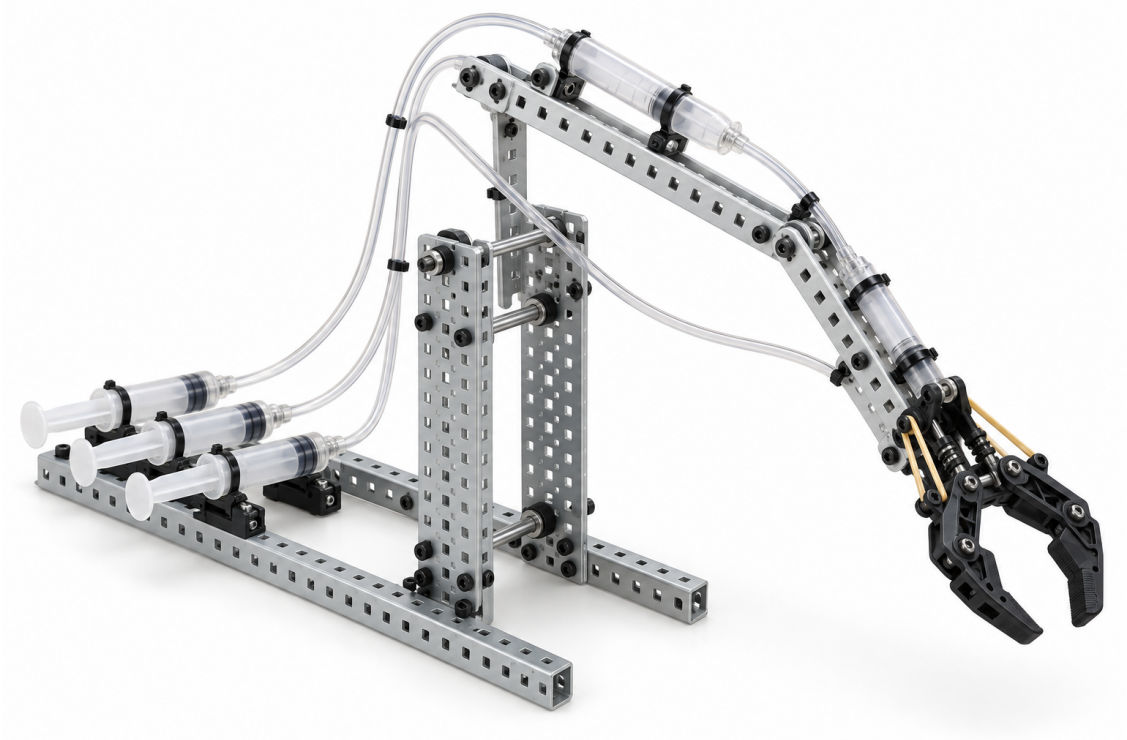
Turn off or unload the mechanism, return reusable components to labeled containers, dry any water system, and record one configuration change plus one observed result. Remove any damaged, loose, frayed, leaking, or hot component from service.

HydroGrab: water-powered tabletop arm

Grades 6-10 | 75-90 min

DRAFT INSTRUCTION MANUAL - RENDERED CONCEPT; PHYSICAL FIRST ARTICLE REQUIRED BEFORE CLASSROOM USE

Build a water-actuated arm and compare predicted with observed force and displacement.



1. Purpose and status

Teams assemble a two-link, syringe-actuated tabletop arm from stock parts only, then use the claw to move a light classroom object. The activity contrasts ideal hydraulic force/displacement relations with loss of accuracy from air, friction, compliance, and linkage angle.

This manual describes an exact-catalog, no-fabrication build sequence. The accompanying render communicates intended arrangement only; it does not verify dimensions, fit, strength, containment, or classroom safety.

2. Safety boundary

Use water, needleless syringes, and lightweight dry objects only. Apply hand force only. Keep all water away from electrical equipment and wipe spills immediately.

Use only factory-provided holes, threads, bores, mounting patterns, and supplied fasteners. Do not cut, drill, punch, mill, saw, bend, heat-form, solder, glue, or 3D-print any component.

3. Before building: inventory and set up

• Count six identical 20 mL needleless syringes. Pair them before filling so each master and slave has the same nominal bore.

• Inspect the Luer ends on each received extension set. Do not force a connection; use the listed female-to-female adapter only when its connector gender is required.

• Arrange a tray, towels, and a dry landing area before filling any circuit.

Pull-list bill of materials

Use the exact vendor item or ASIN shown below. The canonical README in the matching design folder retains the direct procurement hyperlinks and package-yield notes.

Item	Exact purchased configuration and use	Qty. used / purchased
Structural kit	VEX Robotics 276-2161 Metal & Hardware Kit; use the two 5x15 plates, two 1x25 base rails, two 1x25 arm links, supplied #8-32 hardware, bearings, collars, and short shafts	1 / 1 kit
End effector	VEX Robotics 276-6010 Claw Kit v2; use its assembled claw, supplied #64 rubber bands, 12T gear, and 4x-pitch shaft as provided	1 / 1 kit
Syringes	Yalomi ASIN B0DXMZ7XLW, 20 mL Luer-lock syringe without needle, 30-pack; select six identical syringes for three matched master/slave pairs	6 / 30 pack
Fluid lines	ASIN B0GSVQK5WJ, 39 in Luer extension set, 10-pack; use three complete, uncut lines	3 / 10 pack
Luer coupling	ASIN B01KY6X6CS, polypropylene female-to-female Luer-lock adapter, 10-pack; use three only where required to join a line end to a syringe	up to 3 / 10 pack
Retention and strain relief	VEX Robotics 275-0125, 11 in zip ties, 100-pack; secure syringe barrels, plunger-link bars, and tube strain relief	up to 18 / 100 pack
Shared hand tools	VEX Robotics 276-1382 Tool Kit; use the supplied 5/64 in hex key and wrench with the kit hardware	shared / 1 kit

4. Step-by-step assembly

Complete the numbered steps in order. After each mechanical change, move the mechanism by hand with power off and with no payload unless the step explicitly directs otherwise.

5. First-article release checks

A build is not ready for student operation until a facilitator completes and records each applicable check below.

- Every fastener is supplied by 276-2161 and is #8-32; every structural hole used is factory-provided.
- Each water circuit remains sealed under slow hand operation for five cycles; no connector is forced.
- A full syringe stroke does not bottom a structural pivot or side-load a plunger.
- Claw opens/returns without a motor and without a loose part entering a moving linkage.

6. Troubleshooting

Observation	Likely physical cause	First correction
Spongy motion	Trapped air or line compliance	Refill/purge one circuit and use the complete line without a kink
Arm will not lift	Linkage moment demand exceeds available syringe force	Test unloaded; shorten the effective lever arm using another existing VEX hole
Joint sticks	Pivot hardware clamped too tightly or bars rub	Restore washer/bearing clearance and retighten only fixed members
Water leak	Received Luer ends are incompatible or not hand-tight	Stop, dry, verify connector gender, then add the listed female-to-female coupler if needed

Closeout

Turn off or unload the mechanism, return reusable components to labeled containers, dry any water system, and record one configuration change plus one observed result. Remove any damaged, loose, frayed, leaking, or hot component from service.

Build-a-Monocular Lab: threaded 3× Galilean telescope

Grades 8-12 | 45-60 min

DRAFT INSTRUCTION MANUAL - RENDERED CONCEPT; PHYSICAL FIRST ARTICLE REQUIRED BEFORE CLASSROOM USE

Assemble a 3x Galilean monocular and test focus and angular magnification using distant terrestrial targets.



1. Purpose and status

This manual describes an exact-catalog, no-fabrication build sequence. The accompanying render communicates intended arrangement only; it does not verify dimensions, fit, strength, containment, or classroom safety.

2. Safety boundary

Terrestrial viewing only. Never point the instrument at the Sun, a laser, welding arc, or bright specular reflection. Handle every optic by its edge.

Use only factory-provided holes, threads, bores, mounting patterns, and supplied fasteners. Do not cut, drill, punch, mill, saw, bend, heat-form, solder, glue, or 3D-print any component.

3. Before building: inventory and set up

• Use a clean, dry, well-lit table. Remove food, liquids, loose hardware, and abrasive materials from the work area.

• Verify the objective is mounted and the negative eyepiece is free of chips, cracks, fingerprints, and damaged coatings before threading parts together.

• Identify the rotating focus section and its locking ring before installing the eyepiece. Do not use pliers, tape, adhesive, or a cut spacer.

Pull-list bill of materials

Use the exact vendor item or ASIN shown below. The canonical README in the matching design folder retains the direct procurement hyperlinks and package-yield notes.

Function	Exact item and purchased configuration	Buy
Objective	Thorlabs AC254-150-A-ML, Ø1 in mounted achromatic doublet, f=150 mm, 400–700 nm AR coating, SM1-threaded mount	1
Negative eyepiece	Thorlabs LC1715-A, Ø1 in N-BK7 plano-concave lens, f=-50 mm, 350–700 nm AR coating	1
Front spacer tube	Thorlabs SM1L10, SM1 lens tube, 1.0 in nominal thread depth, including one retaining ring	1
Main spacer tube	Thorlabs SM1L20, SM1 lens tube, 2.0 in nominal thread depth, including one retaining ring	1
Fine-focus / eyepiece cell	Thorlabs SM1V05, Ø1 in SM1 rotating adjustable-length lens tube, 0.50 in travel, including one retaining ring and locking ring	1

4. Step-by-step assembly

Complete the numbered steps in order. After each mechanical change, move the mechanism by hand with power off and with no payload unless the step explicitly directs otherwise.

Step 1. Work on a clean, dry table. Confirm that the AC254-150-A-ML, LC1715-A, SM1L10, SM1L20, and SM1V05 are free of chips, cracks, and damaged threads. Handle lenses only by their edges.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 2. Thread the mounted AC254-150-A-ML objective onto the front of SM1L10 by hand. Join SM1L10 to SM1L20, then join SM1L20 to the non-rotating end of SM1V05. Thread parts together only until seated; do not force or cross-thread them.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 3. Seat the LC1715-A in the rear SM1V05 cell with its concave surface facing the objective as the starting configuration. Install the retaining ring supplied with SM1V05 finger-tight, only enough to stop the lens from rattling. Do not use pliers or a sharp tool on the glass.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 4. Point the instrument only at a distant terrestrial target such as a building, tree line, or sign—never at the Sun, a laser, or a bright specular reflection. Look through the eyepiece while slowly rotating the SM1V05 focus section to find the sharpest image, then finger-tighten its locking ring.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 5. Check that the eyepiece cannot move when the monocular is gently tipped. If the image cannot be focused, stop and have the facilitator inspect thread engagement, lens seating, and the first-article optical orientation; do not add a cut spacer or force the focus tube.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

5. First-article release checks

A build is not ready for student operation until a facilitator completes and records each applicable check below.

• Use a distant terrestrial target to verify that a sharp setting exists within the SM1V05 adjustment range and that the locking ring preserves it through handling.

• With the objective capped or pointed at an evenly lit wall, gently tip and rotate the instrument. Verify that neither lens rattles or shifts, and inspect the glass for chips, coating damage, fingerprints, or a loose retaining ring.

• Compare the apparent spacing of two distant vertical features with and without the monocular. The derived target is about 3×; record the measurement method, target distance, observed value, and observer limitations rather than reporting the paraxial value as measured performance.

• Have an adult facilitator inspect each instrument before students use it. Remove any assembly with damaged optics, a loose lens, cross-threaded tubes, or a focus lock that slips.

7. Engineering investigation

For a Galilean monocular, the negative eyepiece intercepts the converging beam from the positive objective. Students can predict M=-f_o/f_e, explain why the image is upright when the ratio is positive, then compare that prediction to an angular-scale observation on a distant terrestrial target. Ask: what changes if the negative eyepiece focal length has a smaller magnitude, and why does the focus setting depend on both the target distance and the observer’s eye?

Closeout

Turn off or unload the mechanism, return reusable components to labeled containers, dry any water system, and record one configuration change plus one observed result. Remove any damaged, loose, frayed, leaking, or hot component from service.

Pinball Lab: reconfigurable tabletop game

Grades 6-12 | 60-90 min

DRAFT INSTRUCTION MANUAL - RENDERED CONCEPT; PHYSICAL FIRST ARTICLE REQUIRED BEFORE CLASSROOM USE

Build a reconfigurable tabletop pinball game and quantify how layout changes affect ball paths and game time.



1. Purpose and status

This manual describes an exact-catalog, no-fabrication build sequence. The accompanying render communicates intended arrangement only; it does not verify dimensions, fit, strength, containment, or classroom safety.

2. Safety boundary

Use one 3/8 in nylon ball per table. Do not release the launcher toward a person. Keep fingers clear of spring hinges while another person launches the ball.

Use only factory-provided holes, threads, bores, mounting patterns, and supplied fasteners. Do not cut, drill, punch, mill, saw, bend, heat-form, solder, glue, or 3D-print any component.

3. Before building: inventory and set up

• Place all perforated plates with their factory hole rows parallel. Confirm the long rails and both end barriers are present before adding the ball.

• Select four straight, undamaged VEX shafts and eight rubber collars for the post bumpers.

• Keep the plunger uncompressed until the launcher adapter is fully bolted and its travel is enclosed by the perimeter rails.

Pull-list bill of materials

Use the exact vendor item or ASIN shown below. The canonical README in the matching design folder retains the direct procurement hyperlinks and package-yield notes.

Function	Exact item and purchased configuration	Buy
Perforated playfield	VEX V5 276-2311, 5 x 25 Aluminum Plate	1 pack
Frame, post shafts, factory-hole end barriers, bearings, standoffs, #8-32 screws/nuts/washers	VEX V5 276-2161, Metal & Hardware Kit	1 kit
Long raised side rails	VEX V5 276-2288, 1x2x1x25 Aluminum C-Channel	1 pack
Cushioned post sleeves	VEX 228-3510, Rubber Shaft Collar (30-pack), for 0.125-in. VEX shafts	1 pack
Play ball	McMaster-Carr 9613K21, 3/8-in. off-white nylon balls (100-pack)	1 pack
Manual launcher	McMaster-Carr 8478A1, nonlocking plate-mount retractable spring plunger	1
Two finger flipper pivots	McMaster-Carr 15205A115, zinc-plated spring hinge, 1-3/4-in. leaves	2
Straight flipper paddles and plunger adapter plates	McMaster-Carr 1555A2, zinc-plated steel flat bracket, 3 in. long, five #8 holes	4
Low-profile target/cushion pads	3M SJ5012BL (Mouser 517-SJ-5012BK), black polyurethane Bumpon pad, 12.7 mm diameter × 3.5 mm high, 56-pad minimum	1 pad
Shared assembly tools	VEX V5 276-1382, Tool Kit	1 shared kit

4. Step-by-step assembly

Complete the numbered steps in order. After each mechanical change, move the mechanism by hand with power off and with no payload unless the step explicitly directs otherwise.

Step 1. Bolt two 276-2311 plates side-by-side with the 276-2161 #8-32 hardware and two supplied standoffs below their seam. Lay the factory hole rows parallel; do not alter the plates.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 2. Attach one full 276-2288 C-channel to each long edge. Bolt one 5x15 VEX steel plate across the upper end and one across the lower end, leaving a controlled ball-return opening beside the lower plate.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 3. Pass four supplied 2-in. VEX shafts through selected square holes in the playfield. Capture each from below with a supplied lock plate. Slide two 228-3510 collars onto each exposed shaft as a soft, fixed post bumper.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 4. Bolt each 15205A115 spring hinge to the plate through its existing holes. Bolt one 1555A2 mending plate to the moving leaf to make each flipper paddle. Confirm that both paddles return under hinge spring force without crossing the ball path.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 5. Use the remaining two 1555A2 brackets as a two-plate adapter for 8478A1; bolt only through existing holes. Install the plunger so its nose points up the playfield and cannot eject the ball over the rail.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 6. Press SJ5012BL pads onto selected flat plate locations as low-impact targets. Put one nylon ball in the launcher lane. Cycle each moving part by hand before playing.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

5. First-article release checks

A build is not ready for student operation until a facilitator completes and records each applicable check below.

• Confirm the hinge and plunger factory-hole patterns can be joined with the listed #8 hardware without drilling or enlarging holes.

• Launch 20 balls at the intended maximum hand force. The ball must remain within the perimeter and return through the lower opening.

• Check that fingers cannot enter the hinge/paddle pinch regions while another student launches a ball.

7. Engineering investigation

The ball’s kinetic energy changes through impacts, rolling resistance, sliding friction, and spring work. A bumper redirects the ball; it does not create energy. Compare bumper-contact count and play time before and after moving one post or pad.

Closeout

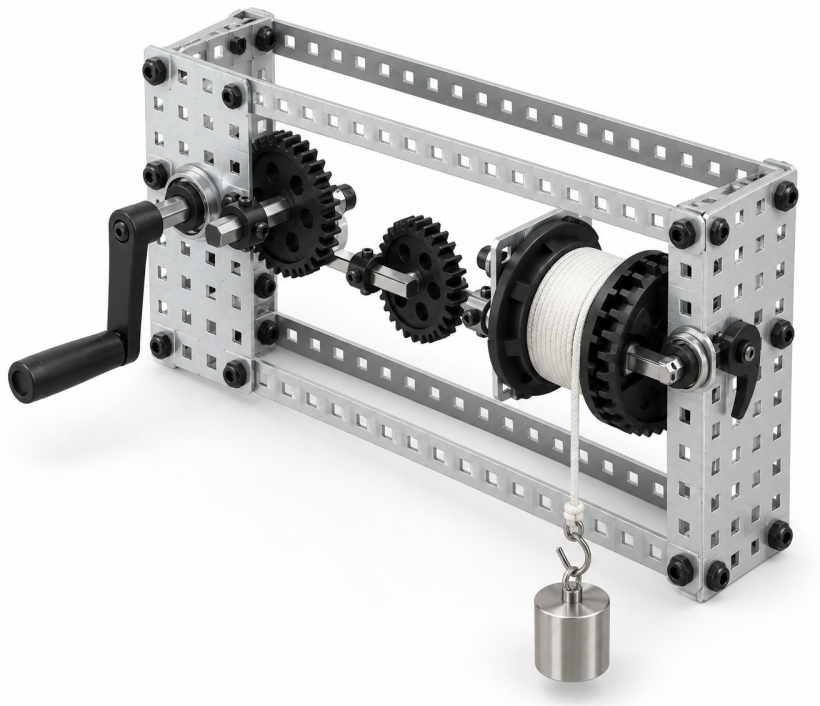
Turn off or unload the mechanism, return reusable components to labeled containers, dry any water system, and record one configuration change plus one observed result. Remove any damaged, loose, frayed, leaking, or hot component from service.

Rescue Hoist: hand-cranked 9:1 gear winch

Grades 7-12 | 90-120 min

DRAFT INSTRUCTION MANUAL - RENDERED CONCEPT; PHYSICAL FIRST ARTICLE REQUIRED BEFORE CLASSROOM USE

Build a two-stage 9:1 hand winch and compare torque reduction with input-turn count.



1. Purpose and status

Teams build a hand-cranked winch with two stock gear stages and a stock ratcheted drum. They first verify the unloaded kinematics, then lift the included 100 g hanging test weight through 20 cm while a spotter maintains the exclusion zone. The investigation compares input turns and perceived effort with direct drive.

This manual describes an exact-catalog, no-fabrication build sequence. The accompanying render communicates intended arrangement only; it does not verify dimensions, fit, strength, containment, or classroom safety.

2. Safety boundary

Hand operation only. Use the specified 100 g hanging test weight and a 20 cm lift above a clear, padded landing surface. Never lift over a person or bypass the ratchet.

Use only factory-provided holes, threads, bores, mounting patterns, and supplied fasteners. Do not cut, drill, punch, mill, saw, bend, heat-form, solder, glue, or 3D-print any component.

3. Before building: inventory and set up

• Mark a no-hand zone below the hanging test weight and around the gear meshes before any rope is installed.

• Set aside two 12T and two 36T gears from the same VEX kit. Do not substitute a 4:1 stage; the intended train is 3:1 times 3:1.

• Inspect the stock rope, drum, pawl, and ratchet. Stop if the rope is frayed or the pawl does not move freely.

Pull-list bill of materials

Use the exact vendor item or ASIN shown below. The canonical README in the matching design folder retains the direct procurement hyperlinks and package-yield notes.

Item	Exact purchased configuration and use	Qty. used / purchased
Frame, shafts, bearings, collars, and fasteners	VEX Robotics 276-2161 Metal & Hardware Kit; use two 5x15 plates, 1x25 base/guard bars, three short square shafts, bearings, collars, and supplied #8-32 hardware	1 / 1 kit
Gear train	VEX Robotics 276-2169 Gear Kit; use two 12T and two 36T 24-DP, 20° gears on the VEX 1/8 in square shaft system	4 / 1 kit
Crank	VEX Robotics 276-2045 Advanced Mechanics and Motion Kit; use its second stock hand crank. The one consolidated kit is shared with Crank Critter, which uses the other hand crank plus the cams/followers.	1 / 1 shared kit
Drum, ratchet, pulleys, and rope	VEX Robotics 276-1546 Winch & Pulley Kit; use one winch bracket/drum/ratchet/pawl and its factory 4 ft nylon rope, uncut	1 / 1 kit
Test load	McMaster-Carr 1914T65 100 g legal-for-trade hanging test weight with top/bottom hooks	1 / 1 each
Shared hand tools	VEX Robotics 276-1382 Tool Kit; use the supplied hex key/wrench with the kit hardware	shared / 1 kit

4. Step-by-step assembly

Complete the numbered steps in order. After each mechanical change, move the mechanism by hand with power off and with no payload unless the step explicitly directs otherwise.

5. First-article release checks

A build is not ready for student operation until a facilitator completes and records each applicable check below.

• Two 12T and two 36T gears are selected from the single 276-2169 kit; the stages are 3:1 × 3:1 = 9:1, not the former 4:1 claim.

• Shafts, bearings, collars, and #8-32 fasteners are from 276-2161; every used shaft fits the VEX square-bore gears.

• The 276-1546 ratchet holds the 1914T65 100 g test weight at rest, and rope never enters a gear mesh.

• A supervisor verifies the gear-access barrier, landing zone, and no-load/100 g sequence before student operation.

6. Troubleshooting

Observation	Likely physical cause	First correction
Teeth climb or skip	Shaft center/parallelism is wrong or mesh is forced	Stop; move to compatible existing holes and realign sidewalls
Handle is hard to turn	Rope rubs, collars are overtight, or a stage binds	Test unloaded; center rope and restore running clearance
Rope piles on one side	Drum/rope approach is misaligned	Reposition the rope guide through an existing hole and rewind by hand
Load descends	Ratchet/pawl is not seated or rope tie is wrong	Lower to landing surface, inspect, and do not resume until pawl engagement is verified

Closeout

Turn off or unload the mechanism, return reusable components to labeled containers, dry any water system, and record one configuration change plus one observed result. Remove any damaged, loose, frayed, leaking, or hot component from service.

Springy Wobblebot: vibrating creature

Grades 5-10 | 30-60 min

DRAFT INSTRUCTION MANUAL - RENDERED CONCEPT; PHYSICAL FIRST ARTICLE REQUIRED BEFORE CLASSROOM USE

Build a low-voltage vibrating creature and investigate how unbalance, compliant legs, and friction produce motion.



1. Purpose and status

This manual describes an exact-catalog, no-fabrication build sequence. The accompanying render communicates intended arrangement only; it does not verify dimensions, fit, strength, containment, or classroom safety.

2. Safety boundary

Use only the listed two-cell AA NiMH supply. Turn power off before any adjustment. Do not use wall power, USB power, alkaline cells, or damaged battery cells.

Use only factory-provided holes, threads, bores, mounting patterns, and supplied fasteners. Do not cut, drill, punch, mill, saw, bend, heat-form, solder, glue, or 3D-print any component.

3. Before building: inventory and set up

• Set the battery holder switch to OFF. Install the two charged cells but do not connect the clip leads yet.

• Choose four identical springs and inspect them for sharp ends, permanent bends, or damage.

• Prepare a clear floor tray for motion tests. Keep hands, hair, clothing, and loose objects outside the eccentric-mass sweep.

Pull-list bill of materials

Use the exact vendor item or ASIN shown below. The canonical README in the matching design folder retains the direct procurement hyperlinks and package-yield notes.

Function	Exact item and purchased configuration	Buy
Body	McMaster-Carr 5257N111, gray impact-resistant polycarbonate enclosure, 3.75 x 4.5 x 2.25 in.	1
Eccentric-mass actuator	Vybronics VZ4TL2B0370055L, 3 V low-current ERM vibration motor with wire leads; supplier series specifies 2.2–3.6 V operation	1
Switched, solder-free power source	Keystone Electronics 2490SW, covered 2xAA holder with on/off switch and 6 in. wire leads	1
Insulated removable connections	Adafruit 4100, 18 in. alligator-to-alligator test-lead set, six leads	1 set
Battery cells	Panasonic eneloop BK-3MCCA4BA, AA NiMH rechargeable batteries, 4-pack, 1.2 V/cell	1 pack
Shared charger for repeat sessions	Panasonic eneloop BQ-CC55SBA, individual-cell AA/AAA NiMH charger	1 shared charger
Four compliant legs	McMaster-Carr 9657K107, zinc-plated spring-tempered-steel compression spring, 1.50 in. long, 0.312 in. OD, 1.1 lbf/in rate, 12-pack	1 pack
Four non-skid feet and motor cushioning	3M SJ5012BL (Mouser 517-SJ-5012BK), black polyurethane Bumpon pad, 12.7 mm diameter × 3.5 mm high, 56-pad minimum	1 pad
Motor, legs, battery holder, and decoration retention	VEX V5 275-0125, 11-in. zip ties (100-pack)	1 pack

4. Step-by-step assembly

Complete the numbered steps in order. After each mechanical change, move the mechanism by hand with power off and with no payload unless the step explicitly directs otherwise.

Step 1. Keep the 2490SW switch off. Install two charged BK-3MCCA4BA cells in the holder, but do not connect the test leads yet. The two-cell nominal supply is 2.4 V and must be recharged only with the shared BQ-CC55SBA NiMH charger.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 2. Use four identical 9657K107 springs from the 12-pack. Use paired 275-0125 cable ties around the enclosure body and the top coil of each spring to retain four legs. Leave each tie full length; do not trim it.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 3. Press one SJ5012BL bumper onto the lower end of each spring. Allow the adhesive to set according to the bumper supplier’s handling guidance before a motion test.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 4. Place two SJ5012BL bumpers on the enclosure top as cushions. Lay VZ4TL2B0370055L across them and use two full 275-0125 cable ties around the enclosure to prevent motor rotation or translation. Ensure its eccentric mass can rotate without touching any tie, wall, or hand.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 5. Retain the battery holder against one enclosure side with two additional full cable ties. Use one red/black pair from 4100: clamp the red lead between the positive holder lead and one motor wire, and the black lead between the negative holder lead and the other motor wire. The exposed clips must not touch each other or any conductive part.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

Step 6. Place the bot on a clear, smooth floor or tray. Turn the switch on for a 5 s supervised trial, then off before changing legs, feet, or cable-tie position.

Verification: Stop before the next step if a fastener will not seat through an existing feature, a moving part binds, a rope/tube/wire rubs, or a component requires force beyond normal hand assembly.

5. First-article release checks

A build is not ready for student operation until a facilitator completes and records each applicable check below.

• Use only the specified two-cell BK-3MCCA4BA NiMH arrangement (2.4 V nominal), which lies within the supplier-stated 2.2–3.6 V operating range of the selected vibration-motor series. This design excludes USB supplies, wall supplies, alkaline cells, and battery types other than the specified 2xAA NiMH arrangement.

• With power off, rotate the eccentric mass by hand; it must clear the enclosure, ties, clips, springs, and student-accessible edges.

• Run five 10 s trials on a floor tray. Inspect cable-tie retention, bumper adhesion, clip grip, wire insulation, and enclosure temperature after each trial. Remove the build from use if a clip loosens, a wire is damaged, or the motor is hot to touch.

7. Engineering investigation

The rotating offset mass creates a periodic force with magnitude that scales approximately as Fapprox m e omega^2, where m is eccentric mass, e is offset, and omega is angular speed. Motion across the floor depends on friction and how vibration couples to the spring legs; it need not increase monotonically with offset.

Closeout

Turn off or unload the mechanism, return reusable components to labeled containers, dry any water system, and record one configuration change plus one observed result. Remove any damaged, loose, frayed, leaking, or hot component from service.